

Lab-5 Ex3 - Factorial

In the lecture we mentioned that sometimes we use the loop variable directly in our computation. In these cases, we need to be very aware of how our loop variable changes from its initial value to its terminating value.

In this exercise, let's try to write a C program that calculates a factorial. For a positive integer x , the factorial of x , expressed in $x!$, is given by:

$$x! = (x) * (x-1) * \dots * 1$$

To use looping to solve the problem, we can set x to be our loop variable, and decrease it by 1 every time until x reaches the value of 1.

Please follow the following **pseudocode** to complete the program:

```
define integer x

obtain integer x from user

define integer factorial = 1

while x >= 1

    factorial = factorial * x

    x = x - 1

end of loop

print factorial
```

Assume the input is in range of 1 to 15. User-input integers are ***bold and italic*** in the following sample runs.

Sample run #1:

5
120

Sample run #2:

15
2004310016

Sample run #3:

1

*P.S. Do you know why we restrict the range of input x ? Use a calculator to get the answer for **15!**. Do you think we should restrict the range further?*

(NOTE: Please follow the pseudocode to complete the program anyhow)